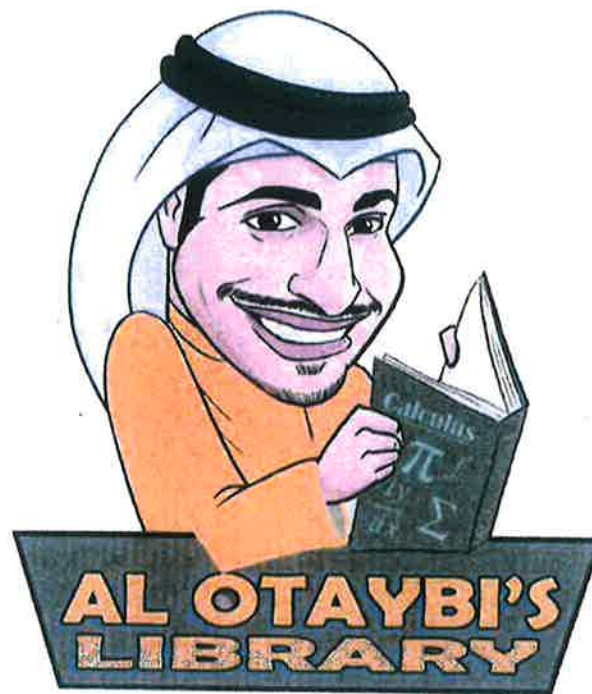


Kuwait University

2021/2022

Spring course



Calculus B

First

Section : 7.1

7.1 Integration by Parts.

(3)



Quiz #2



$$\int \left(\frac{1}{2}x + e^{\frac{1}{2}x} + \left(\frac{1}{2}\right)^x \right) dx$$

$$\int \frac{2}{\sqrt{9-x^2}} dx$$

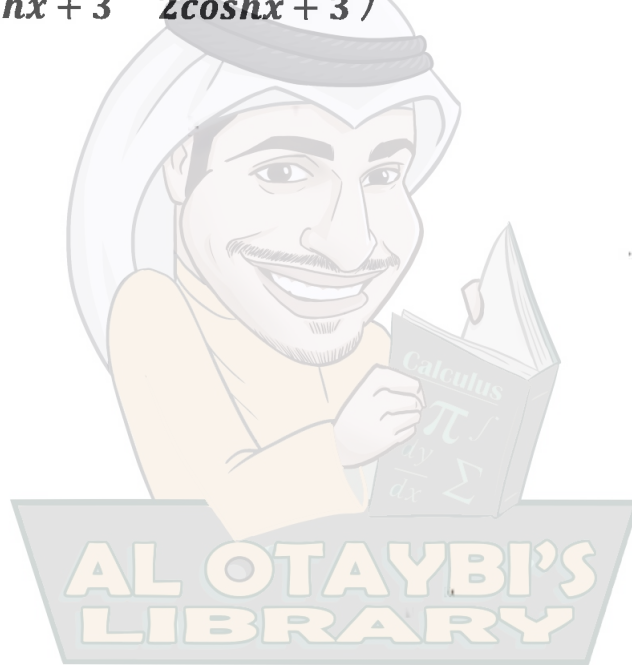
$$\int \frac{2x}{\sqrt{9-x^2}} dx$$

$$\int \frac{3^{\ln x}}{x} dx$$

$$\int x^2 \cot(x^3 + 2) dx$$

$$\int \left(\frac{\sinh x}{\cosh x + 3} - \frac{2\sinh x}{\cosh x + 3} + \frac{\sinh x}{2\cosh x + 3} \right) dx$$

$$\int \frac{\sinh x}{\cosh^2 x + 3} dx$$





7.1 Integration By parts



طريقة يمكن من خلالها تحويل التكاملات المعقدة الى تكاملات سهلة يمكن التعامل معها

u

$\ln$

Inverse of trigonometric function
($\sin^{-1}x - \cos^{-1}x - \tan^{-1}x \dots$)

dv

$e^x - a^x$

مشتقات الدوال المثلثية و مشتقات (h)
($\sin x - \cos x - \sec^2 x \dots$)

خطوات الحل



u

و

dv

(1) تحديد



اشتقاق

تكامل

(2)

AL OTAYBI'S
LIBRARY

(3) تطبيق القانون

$$u v - \int v du$$

مثال توضيحي

$$\int x e^x dx$$



تكامل معقد

$$u = x$$

أشتق

$$du = dx$$

$$dv = e^x dx$$

أكامل

$$v = e^x$$

لازم السؤال كله يفرض
u و dv



$$\begin{aligned} uv - \int v du \\ x e^x - \int e^x dx \\ x e^x - e^x + C \end{aligned}$$



تكامل سهل
قانون

AL OTAYBI'S
LIBRARY

$$\int x \sec^2 x \, dx$$

$$u = x \quad dv = \sec^2 x \, dx$$

$$du = dx \quad \leftarrow \rightarrow \quad v = \tan x$$

$$uv - \int v \, du$$

$$x \tan x - \int \tan x \, dx$$

$$x \tan x - \ln |\sec x| + C$$

$$\int x \sec x \tan x \, dx$$

$$u = x \quad dv = \sec x \tan x \, dx$$

$$du = dx \quad \leftarrow \rightarrow \quad v = \sec x$$

$$uv - \int v \, du$$

$$x \sec x - \int \sec x \, dx$$

$$x \sec x - \ln |\sec x + \tan x| + C$$



Text Book Questions



$$\int x \cos 5x \, dx$$

$$u = x \quad dv = \cos 5x \, dx$$
$$du = dx \quad v = \frac{\sin 5x}{5}$$

$$x \frac{\sin 5x}{5} - \int \frac{\sin 5x}{5} \, dx$$
$$\frac{1}{5} x \sin 5x - \frac{1}{5} \int \sin 5x \, dx$$
$$\frac{1}{5} x \sin 5x - \frac{1}{5} \left(-\frac{\cos 5x}{5} \right) + C$$

$$\int \sqrt{x} \ln x \, dx$$

$$u = \ln x \quad dv = \sqrt{x} \, dx$$
$$du = \frac{1}{x} \, dx \quad v = \frac{2}{3} x^{\frac{3}{2}}$$

$$\frac{2}{3} \ln x \cdot x^{\frac{3}{2}} - \frac{2}{3} \int x^{\frac{3}{2}} \frac{1}{x} \, dx$$
$$\frac{2}{3} \ln x \cdot x^{\frac{3}{2}} - \frac{2}{3} \int x^{\frac{1}{2}} \, dx$$
$$\frac{2}{3} \ln x \cdot x^{\frac{3}{2}} - \frac{2}{3} \cdot \frac{2}{\frac{3}{2}} x^{\frac{3}{2}} + C$$

$$\int x 3^x dx$$

$$u = x \quad dv = 3^x$$

$$du = dx \quad v = \frac{3^x}{\ln 3}$$



إذا كان أحد السهمين فيه رقم
أطلعه برا ←

$$\frac{1}{\ln 3} x 3^x - \frac{1}{\ln 3} \int 3^x dx$$

$$\frac{1}{\ln 3} x 3^x - \frac{1}{\ln 3} \frac{3^x}{\ln 3} + C$$

$$\int x \frac{\cos x}{\sin^2 x} dx$$

$$\int x \cdot \frac{\cos x}{\sin x \sin x} dx$$

$$\int x \csc x \cot x dx$$

$$u = x \quad dv = \csc x \cot x dx$$

$$du = dx \quad v = -\csc x$$

$$- x \csc x + \int \csc x dx$$

$$- x \csc x + \ln |\csc x - \cot x| + C$$

$$\int \frac{\ln x}{x^3} dx$$

$$u = \ln x \quad dv = \frac{1}{x^3} dx$$

$$du = \frac{1}{x} dx \quad v = \frac{x^{-2}}{-2}$$

$$-\frac{1}{2} \ln x x^{-2} + \frac{1}{2} \int x^{-2} \frac{1}{x} dx$$

$$-\frac{1}{2} \ln x \cdot x^{-2} + \frac{1}{2} \int x^{-3} dx$$

$$-\frac{1}{2} \ln x \cdot x^{-2} + \frac{1}{2} \frac{x^{-2}}{-2} + C$$

$$\int \frac{\ln \sqrt{x}}{x^3} dx$$

$$\int \frac{\ln x^{\frac{1}{2}}}{x^3} dx$$

يمكن التبسيط
لجواب
ln

$$\frac{1}{2} \int \frac{\ln x}{x^3} dx$$



$$u = \ln x \quad dv = \frac{1}{x^3} dx$$

$$du = \frac{1}{x} dx \quad v = \frac{x^{-2}}{-2}$$

$$\frac{1}{2} \left[-\frac{1}{2} \ln x x^{-2} + \frac{1}{2} \int x^{-2} \frac{1}{x} dx \right]$$

$$\frac{1}{2} \left[-\frac{1}{2} \ln x x^{-2} + \frac{1}{2} \frac{x^{-2}}{-2} \right] + C$$



Text Book Questions



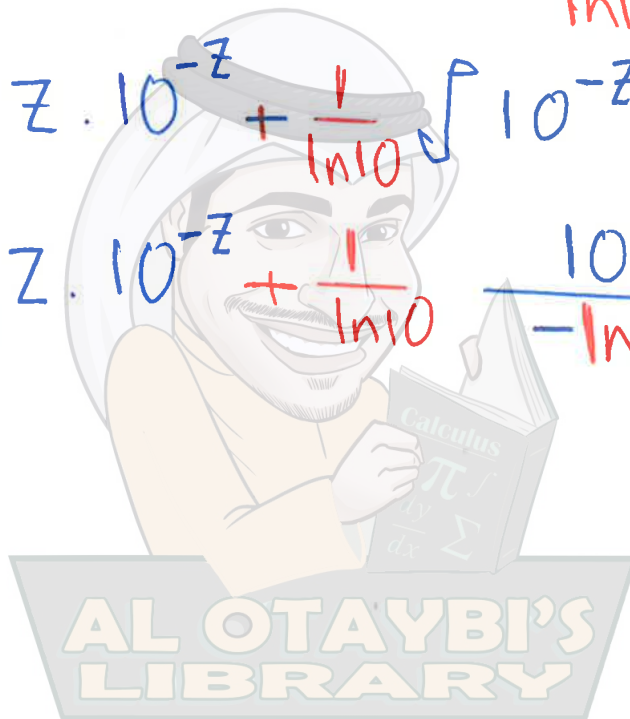
$$\int \frac{z}{10^z} dz$$

$$\int z 10^{-z} dz$$

$$u = z \quad dv = 10^{-z} dz$$
$$du = dz \quad v = \frac{10^{-z}}{-\ln 10}$$

$$\frac{-1}{\ln 10} z \cdot 10^{-z} + \frac{1}{\ln 10} \int 10^{-z} dz$$

$$\frac{-1}{\ln 10} z \cdot 10^{-z} + \frac{1}{\ln 10} \frac{10^{-z}}{-\ln 10} + C$$





Text Book Questions



$$\int t^4 \ln t \, dt$$

$$\begin{aligned} u &= \ln t & dv &= t^4 \, dt \\ du &= \frac{1}{t} \, dt & v &= \frac{t^5}{5} \\ \frac{1}{5} \ln t \cdot t^5 - \frac{1}{5} \int t^5 \frac{1}{t} \, dt \\ \frac{1}{5} \ln t \cdot t^5 - \frac{1}{5} \int t^4 \, dt \\ \frac{1}{5} \ln t \cdot t^5 - \frac{1}{5} \frac{t^5}{5} + C \end{aligned}$$

$$\int (x^2 + 2x) \cos x \, dx$$

$$\begin{aligned} u &= x^2 + 2x & dv &= \cos x \, dx \\ du &= (2x + 2) \, dx & v &= \sin x \\ (x^2 + 2x) \sin x - \int \sin x (2x + 2) \, dx \end{aligned}$$

$$\begin{aligned} u &= 2x + 2 & dv &= \sin x \, dx \\ du &= 2 \, dx & v &= -\cos x \\ -(2x + 2) \cos x + 2 \int \cos x \, dx \\ -(2x + 2) \cos x + 2 \sin x + C \\ (x^2 + 2x) \sin x - [- (2x + 2) \cos x + 2 \sin x] + C \end{aligned}$$

$$\int x(\ln x + e^x) dx$$

$$\int (x \ln x + x e^x) dx$$

$$\int x \ln x dx + \int x e^x dx$$

$$u = \ln x \quad dv = x dx$$

$$du = \frac{1}{x} dx \quad v = \frac{x^2}{2}$$

$$u = x \quad dv = e^x dx$$

$$du = dx \quad v = e^x$$

$$\frac{1}{2} \ln x \cdot x^2 - \frac{1}{2} \int x^2 \frac{1}{x} dx$$

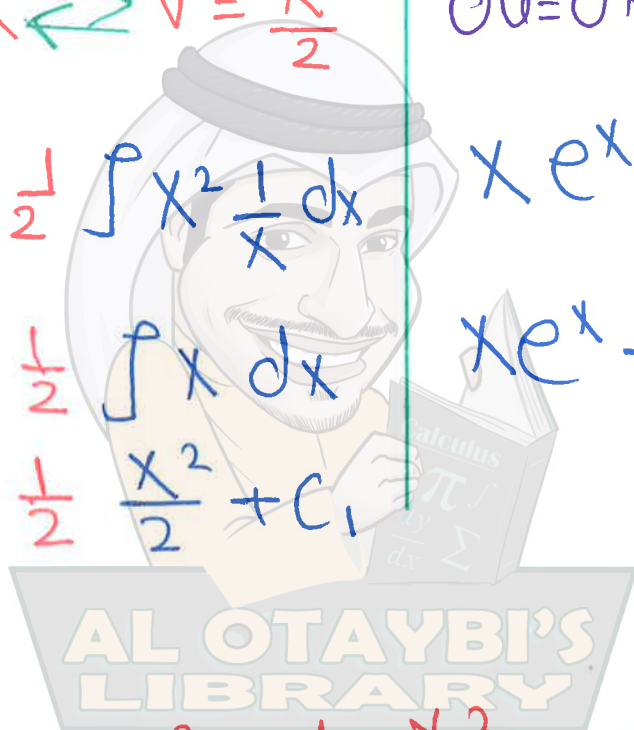
$$x e^x - \int e^x dx$$

$$\frac{1}{2} \ln x \cdot x^2 - \frac{1}{2} \int x dx$$

$$x e^x - e^x + C_2$$

$$\frac{1}{2} \ln x \cdot x^2 - \frac{1}{2} \frac{x^2}{2} + C_1$$

$$\Rightarrow \frac{1}{2} \ln x \cdot x^2 - \frac{1}{2} \frac{x^2}{2} + x e^x - e^x + C$$



$$\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$$

$$u = \sin^{-1} x$$

$$dv = \frac{x}{\sqrt{1-x^2}} dx$$

$$du = \frac{1}{\sqrt{1-x^2}} dx$$

$$v = \int \frac{x}{\sqrt{1-x^2}} dx$$

$$= \int x(1-x^2)^{-\frac{1}{2}} dx$$

$$\text{let } t = 1-x^2$$

$$dt = -2x dx$$

$$-\frac{1}{2} dt = x dx$$

$$= -\frac{1}{2} \int t^{-\frac{1}{2}} dt$$

$$= -\frac{1}{2} \frac{t^{\frac{1}{2}}}{\frac{1}{2}}$$

$$= -\sqrt{t}$$

$$= -\sqrt{1-x^2}$$

$$\begin{aligned} & - \sin^{-1} x \sqrt{1-x^2} + \int \cancel{\sqrt{1-x^2}} \frac{1}{\cancel{\sqrt{1-x^2}}} dx \\ & - \sin^{-1} x \sqrt{1-x^2} + \int dx \\ & - \sin^{-1} x \sqrt{1-x^2} + x + C \end{aligned}$$



$$\int x \sec^{-1} x \, dx$$

$$u = \sec^{-1} x \quad dv = x \, dx$$

$$du = \frac{1}{x \sqrt{x^2 - 1}} dx \quad v = \frac{x^2}{2}$$

$$\frac{1}{2} \sec^{-1} x \cdot x^2 - \frac{1}{2} \int x^2 \frac{1}{x \sqrt{x^2 - 1}} dx$$

$$\int \frac{x}{\sqrt{x^2 - 1}} dx$$

$$\int x (x^2 - 1)^{-\frac{1}{2}} dx$$

$$\text{Let } t = x^2 - 1$$

$$dt = 2x \, dx$$

$$\frac{1}{2} dt = x \, dx$$

$$\frac{1}{2} \int t^{-\frac{1}{2}} dt$$

$$\frac{1}{2} \frac{t^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$(x^2 - 1)^{\frac{1}{2}} + C$$

$$\Rightarrow \frac{1}{2} \sec^{-1} x \cdot x^2 - \frac{1}{2} (x^2 - 1)^{\frac{1}{2}} + C$$



Text Book Question



$$\int \arctan\left(\frac{1}{x}\right) dx$$

$$\int \tan^{-1}\left(\frac{1}{x}\right) dx$$

$$u = \tan^{-1}\left(\frac{1}{x}\right) \quad dv = dx$$

$$\begin{aligned} du &= \frac{-\frac{1}{x^2}}{\left(\frac{1}{x}\right)^2 + 1} dx & v &= x \\ &= \frac{-\frac{1}{x^2}}{\frac{1}{x^2} + \frac{x^2}{x^2}} dx \\ &= \frac{-\frac{1}{x^2}}{1 + x^2} dx \\ &= \frac{-1}{1 + x^2} dx \end{aligned}$$

$$\begin{aligned} & \tan^{-1}\left(\frac{1}{x}\right) \cdot x + \int \frac{x}{1 + x^2} dx \\ & \quad \left(\frac{1}{2}\right) \int \frac{2x}{1 + x^2} dx \\ & \tan^{-1}\left(\frac{1}{x}\right)x + \frac{1}{2} \ln|1 + x^2| + C \end{aligned}$$



Text Book Questions



$$\int x \ln(x+1) dx$$

$$u = \ln(x+1) \quad dv = x dx$$

$$du = \frac{1}{x+1} dx \quad \rightarrow \quad v = \frac{x^2}{2}$$

$$\frac{1}{2} \ln(x+1) \cdot x^2 - \frac{1}{2} \int \frac{x^2}{x+1} dx$$



دائما اذا عندي

(1) $\frac{\text{حدودية}}{\text{حدودية}}$

(2) البسط $\leq$ المقام

قسمة مطولة

$$\begin{array}{r} x-1 \\ x^2 \\ \hline 1+x \overline{) x^2 - x} \\ \underline{-(x^2 + x)} \\ -x \\ \underline{-(x + 1)} \\ 1 \end{array}$$

$$\int \left[x-1 + \frac{1}{x+1} \right] dx$$

$$= \frac{x^2}{2} - x + \ln|x+1| + C$$

$$\frac{1}{2} \ln(x+1) \cdot x^2 - \frac{1}{2} \left[\frac{x^2}{2} - x + \ln|x+1| \right] + C$$

$$\int x^2 \tan^{-1} x \, dx$$

$$u = \tan^{-1} x \quad dv = x^2 dx$$

$$du = \frac{1}{x^2 + 1} dx \quad v = \frac{x^3}{3}$$

$$\frac{1}{3} \tan^{-1} x \cdot x^3 - \frac{1}{3} \int \frac{x^3}{x^2 + 1} dx$$



دائما اذا عندي

(1) $\frac{\text{حدودية}}{\text{حدودية}}$

(2) البسط $\leq$ المقام

قسمة مطولة

$$\int \frac{x^3}{x^2 + 1} dx$$

$$\begin{array}{r} x \\ 1 + x^2 \overline{) x^3} \\ \underline{-x^3} \\ 0 \end{array}$$

$$\int \left[\frac{x^2}{x^2 + 1} + \frac{-x}{x^2 + 1} \right] dx$$

$$\frac{x^2}{2} - \left(\frac{1}{2} \right) \int \frac{2x}{x^2 + 1} dx$$

$$\frac{x^2}{2} - \frac{1}{2} \ln|x^2 + 1| + C$$

$$\Rightarrow \frac{1}{3} \tan^{-1} x \cdot x^3 - \frac{1}{3} \left[\frac{x^2}{2} - \frac{1}{2} \ln|x^2 + 1| \right] + C$$

$$\int x^2 \sinh x \, dx$$

$$u = x^2 \quad dv = \sinh x \, dx$$

$$du = 2x \, dx \quad v = \cosh x$$

$$x^2 \cosh x - 2 \int \cosh x \cdot x \, dx$$

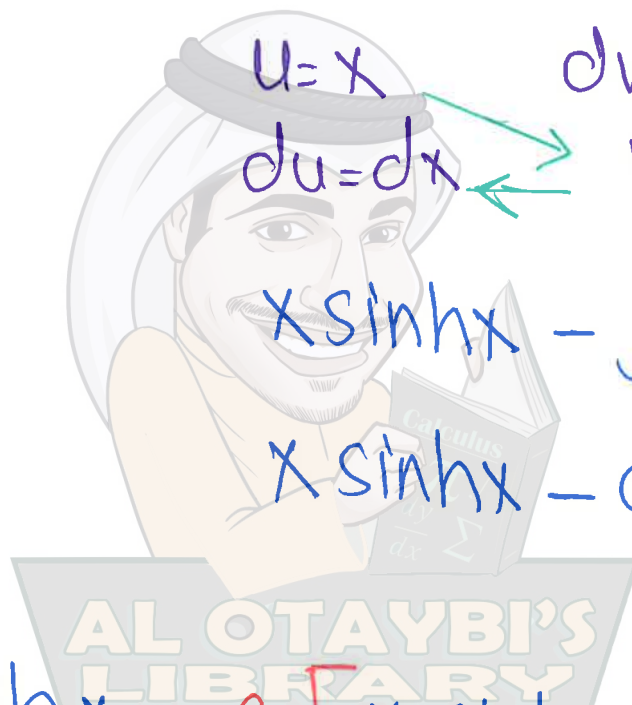
$$\int \cosh x \cdot x \, dx$$

$$u = x \quad dv = \cosh x \, dx$$

$$du = dx \quad v = \sinh x$$

$$x \sinh x - \int \sinh x \, dx$$

$$x \sinh x - \cosh x + C$$



$$\Rightarrow x^2 \cosh x - 2 [x \sinh x - \cosh x] + C$$

$$\int x^2 4^x dx$$

Nov 2016

$$u = x^2 \quad dv = 4^x dx$$

$$du = 2x dx \quad \longleftrightarrow \quad v = \frac{4^x}{\ln 4}$$

$$\frac{1}{\ln 4} x^2 4^x - \frac{2}{\ln 4} \int 4^x \cdot x dx$$

$$u = x \quad dv = 4^x dx$$

$$du = dx \quad \longleftrightarrow \quad v = \frac{4^x}{\ln 4}$$

$$\frac{1}{\ln 4} x \cdot 4^x - \frac{1}{\ln 4} \int 4^x dx$$

$$\frac{1}{\ln 4} x \cdot 4^x - \frac{1}{\ln 4} \frac{4^x}{\ln 4} + C$$

$$\frac{1}{\ln 4} x^2 4^x - \frac{2}{\ln 4} \left[\frac{1}{\ln 4} x \cdot 4^x - \frac{1}{\ln 4} \frac{4^x}{\ln 4} \right] + C$$



Text Book Question



$$\int_4^9 \frac{\ln y}{\sqrt{y}} dy$$

$$u = \ln y \quad dv = \frac{1}{\sqrt{y}} dy$$

$$du = \frac{1}{y} dy \quad v = 2 y^{\frac{1}{2}}$$

$$2 \ln y \cdot y^{\frac{1}{2}} - 2 \int y^{\frac{1}{2}} \frac{1}{y} dy$$

$$\left[2 \ln y \cdot y^{\frac{1}{2}} - 2 \cdot 2 y^{\frac{1}{2}} \right]_4^9$$

$$2 \ln 9 (3) - 4(3) - (2 \ln 4 (2) - 4(2))$$

$$6 \ln 9 - 12 - 4 \ln 4 + 8$$

$$6 \ln 9 - 4 \ln 4 - 4$$



Text Book Question



$$\int_1^2 \frac{(\ln x)^2}{x^3} dx$$

$$u = (\ln x)^2$$

$$dv = \frac{1}{x^3} dx$$

x^{-3}

$$du = 2(\ln x) \frac{1}{x} dx$$

$$v = \frac{x^{-2}}{-2}$$

$$-\frac{1}{2} (\ln x)^2 \cdot x^{-2} + \frac{2}{2} \int x^{-2} (\ln x) \frac{1}{x} dx$$

$$\int x^{-3} \ln x dx$$

$$u = \ln x$$

$$dv = x^{-3} dx$$

$$du = \frac{1}{x} dx$$

$$v = \frac{x^{-2}}{-2}$$

$$-\frac{1}{2} \ln x \cdot x^{-2} + \frac{1}{2} \int x^{-2} \frac{1}{x} dx$$

x^{-3}

$$-\frac{1}{2} \ln x \cdot x^{-2} + \frac{1}{2} \frac{x^{-2}}{-2}$$

$$\left[-\frac{1}{2} (\ln x)^2 \frac{1}{x^2} - \frac{1}{2} \ln x \cdot \frac{1}{x^2} - \frac{1}{4} \frac{1}{x^2} \right]_1^2$$

$$\left(-\frac{1}{2} (\ln 2)^2 \frac{1}{4} - \frac{1}{2} \ln 2 \cdot \frac{1}{4} - \frac{1}{4} \frac{1}{4} \right) - \left(0 - 0 - \frac{1}{4} \right)$$



Note



$$\int e^x dx = e^x + c$$



أولوية **dv** بروحها
قانون

$$\int \ln x dx = ??!!$$



أولوية **u** بروحها
By parts

$$\int (\ln x)^2 dx$$

$$u = (\ln x)^2 \quad dv = dx$$

$$du = 2(\ln x) \cdot \frac{1}{x} dx \quad v = x$$

$$(\ln x)^2 \cdot x - 2 \int x (\ln x) \frac{1}{x} dx$$

$$\int \ln x dx$$

$$u = \ln x \quad dv = dx$$

$$du = \frac{1}{x} dx \quad v = x$$

$$\ln x \cdot x - \int x \frac{1}{x} dx$$

$$\ln x \cdot x - x + c$$

$$\Rightarrow (\ln x)^2 x - 2 [\ln x \cdot x - x] + c$$

$$\int (\log_3 x)^2 dx$$

Past exam

$$\int \left(\frac{1}{\ln 3} \ln x \right)^2 dx = \int \frac{1}{(\ln 3)^2} (\ln x)^2 dx$$

$$\frac{1}{(\ln 3)^2} \int (\ln x)^2 dx$$

$$u = (\ln x)^2$$

$$dv = dx$$

$$du = 2(\ln x) \cdot \frac{1}{x} dx$$

$$v = x$$

$$\frac{1}{(\ln 3)^2} \left[(\ln x)^2 \cdot x - 2 \int x \ln x \cdot \frac{1}{x} dx \right]$$

$$u = \ln x$$

$$dv = dx$$

$$du = \frac{1}{x} dx$$

$$v = x$$

$$\ln x \cdot x - \int x \cdot \frac{1}{x} dx$$

$$\ln x \cdot x - x + C$$

$$\frac{1}{(\ln 3)^2} \left[(\ln x)^2 \cdot x - 2 (\ln x \cdot x - x) \right] + C$$



Text Book Questions



$$\int \cos^{-1} x \, dx$$

$$u = \cos^{-1} x \quad dv = dx$$
$$du = \frac{-1}{\sqrt{1-x^2}} dx \quad v = x$$

$$\cos^{-1} x \cdot x + \int \frac{x}{\sqrt{1-x^2}} dx$$

$$\int x (1-x^2)^{-\frac{1}{2}} dx$$

$$\text{Let } t = 1-x^2$$

$$dt = -2x \, dx$$

$$\frac{1}{2} dt = -x \, dx$$

$$-\frac{1}{2} \int t^{-\frac{1}{2}} dt$$

$$-\frac{1}{2} \frac{t^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$-\sqrt{t} + C$$

$$-\sqrt{1-x^2} + C$$

$$\cos^{-1} x \cdot x - \sqrt{1-x^2} + C$$

Use integration by parts to prove the reduction formula

$$\int x^n e^x dx = x^n e^x - n \int x^{n-1} e^x dx$$

$$\text{L.H.S} = \int x^n e^x dx$$

$$u = x^n \quad dv = e^x dx$$

$$du = nx^{n-1} dx \quad \swarrow \searrow \quad v = e^x$$

$$x^n e^x - n \int e^x x^{n-1} dx$$

$$= \text{R.H.S}$$

Use integration by parts to prove the reduction formula

$$\int (\ln x)^n dx = x(\ln x)^n - n \int (\ln x)^{n-1} dx$$

$$\text{L.H.S} = \int (\ln x)^n dx$$

$$u = (\ln x)^n \quad dv = dx$$

$$du = n(\ln x)^{n-1} \frac{1}{x} dx \quad \swarrow \searrow \quad v = x$$

$$(\ln x)^n \cdot x - n \int x(\ln x)^{n-1} \frac{1}{x} dx$$

$$(\ln x)^n x - n \int (\ln x)^{n-1} dx$$

$$= \text{R.H.S}$$



أولويات منخشة !!



$$\tan^2 x$$

$$\cot^2 x$$

قوانين التحويل

$$1 + \tan^2 x = \sec^2 x$$

$$\cot^2 x + 1 = \csc^2 x$$

$$\sin^2 x$$

$$\cos^2 x$$

قوانين الدوا

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\int x \tan^2 x \, dx$$

$$\int x (\sec^2 x - 1) \, dx$$

$$\int (x \sec^2 x - x) \, dx$$

$$\int x \sec^2 x \, dx - \int x \, dx$$

$$u = x \quad dv = \sec^2 x \, dx$$

$$du = dx \quad v = \tan x$$

$$\frac{x^2}{2} + C_2$$

$$x \tan x - \int \tan x \, dx$$

$$x \tan x - \ln |\sec x| + C_1$$

$$x \tan x - \ln |\sec x| - \frac{x^2}{2} + C$$

$$\int x \sin^2 x \, dx$$

Oct 2017

$$\int x \cdot \frac{1}{2} (1 - \cos 2x) \, dx$$

$$\frac{1}{2} \int x (1 - \cos 2x) \, dx$$

$$\frac{1}{2} \int (x - x \cos 2x) \, dx$$

$$\frac{1}{2} \left[\int x \, dx - \int x \cos 2x \, dx \right]$$

$$\frac{x^2}{2} + C_1$$

$$\int x \cos 2x \, dx$$

$$u = x$$

$$du = dx$$

$$dv = \cos 2x \, dx$$

$$v = \frac{\sin 2x}{2}$$

$$\frac{1}{2} x \sin 2x - \frac{1}{2} \int \sin 2x \, dx$$

$$\frac{1}{2} x \sin 2x - \frac{1}{2} \left(\frac{-\cos 2x}{2} \right) + C_2$$

$$\frac{1}{2} \left[\frac{x^2}{2} - \left[\frac{1}{2} x \sin 2x - \frac{1}{2} \left(\frac{-\cos 2x}{2} \right) \right] \right] + C$$



Multiple choice



$\int te^{2t} dt$ equals

Oct 2017

(a) $\frac{1}{4}(2t+2)e^{2t} + C$ (b) $\frac{1}{4}(t+1)e^{2t} + C$

(c) $\frac{1}{4}(2t+1)e^{2t} + C$ (d) $\frac{1}{2}(2t+1)e^{2t} + C$

(e) $\frac{1}{4}(2t-1)e^{2t} + C$

$\int te^{2t} dt$

$u = t$
 $du = dt$

$dv = e^{2t} dt$
 $v = \frac{e^{2t}}{2}$

$\frac{1}{2} te^{2t} - \frac{1}{2} \int e^{2t} dt$

$\frac{1}{2} te^{2t} - \frac{1}{2} \frac{e^{2t}}{2} + C$

$\frac{1}{2} te^{2t} - \frac{1}{4} e^{2t} + C$

$\frac{1}{4} e^{2t} (2t - 1) + C$ (e)



التكامل اللولبي



$$\int e^x \cos(3x) dx = I$$

$$u = \cos(3x) \quad \rightarrow \quad dv = e^x dx$$

$$du = -3 \sin(3x) dx \quad \leftarrow \quad v = e^x$$

$$\cos(3x) e^x + 3 \int e^x \sin(3x) dx$$

$$\int e^x \sin(3x) dx$$

$$u = \sin(3x) \quad dv = e^x dx$$

$$du = 3 \cos(3x) dx \quad \leftarrow \quad v = e^x$$

$$\sin(3x) e^x - 3 \int e^x \cos(3x) dx$$

$$I = \cos(3x) e^x + 3 [\sin(3x) e^x - 3 I]$$

$$I = \cos(3x) e^x + 3 \sin(3x) e^x - 9 I$$

$$10 I = \cos(3x) e^x + 3 \sin(3x) e^x$$

$$I = \frac{1}{10} [\cos(3x) e^x + 3 \sin(3x) e^x] + C$$

$$\int 2^x \cos x \, dx = I$$

$$u = 2^x \quad dv = \cos x \, dx$$

$$du = 2^x \ln 2 \, dx \quad v = \sin x$$

$$2^x \sin x - \ln 2 \int \sin x \cdot 2^x \, dx$$

$$\int \sin x \cdot 2^x \, dx$$

$$u = 2^x \quad dv = \sin x \, dx$$

$$du = 2^x \ln 2 \, dx \quad v = -\cos x$$

$$-2^x \cos x + \ln 2 \int \cos x \cdot 2^x \, dx$$

$$I = 2^x \sin x - \ln 2 \left[-2^x \cos x + \ln 2 I \right]$$

$$I = 2^x \sin x + \ln 2 \cdot 2^x \cos x - (\ln 2)^2 I$$

$$I + (\ln 2)^2 I = 2^x \sin x + \ln 2 \cdot 2^x \cos x$$

$$I [1 + (\ln 2)^2] = 2^x \sin x + \ln 2 \cdot 2^x \cos x$$

$$I = \frac{1}{1 + (\ln 2)^2} [2^x \sin x + \ln 2 \cdot 2^x \cos x]$$

Text book
question

$$\int \cos(\ln x) dx = I$$

$$u = \cos(\ln x) \quad dv = dx$$

$$du = -\sin(\ln x) \frac{1}{x} dx \quad v = x$$

$$\cos(\ln x) \cdot x + \int x \sin(\ln x) \frac{1}{x} dx$$
$$\int \sin(\ln x) dx$$

$$u = \sin(\ln x) \quad dv = dx$$

$$du = \cos(\ln x) \frac{1}{x} dx \quad v = x$$

$$\sin(\ln x) x - \int x \cos(\ln x) \frac{1}{x} dx$$

$$I = \cos(\ln x) x + \sin(\ln x) x - I$$

$$2 I = \cos(\ln x) x + \sin(\ln x) x$$

$$I = \frac{1}{2} [\cos(\ln x) x + \sin(\ln x) x] + C$$



Try to solve !!



$$\int \cos(\ln x^2) dx$$

$$\int \cos(2 \ln x) dx$$

$$u = \cos(2 \ln x)$$
$$du = -\sin(2 \ln x) \frac{2}{x} dx$$

$$dv = dx$$

$$v = x$$

المحل

$$\boxed{\int \sin(\ln \sqrt{x}) dx} = I$$

$$\int \sin(\ln x^{\frac{1}{2}}) dx$$

$$\int \sin\left(\frac{1}{2} \ln x\right) dx$$

$$u = \sin\left(\frac{1}{2} \ln x\right) \quad dv = dx$$

$$du = \cos\left(\frac{1}{2} \ln x\right) \cdot \frac{1}{2} \cdot \frac{1}{x} dx \quad \leftarrow \begin{matrix} \nearrow \\ \nwarrow \end{matrix} \quad v = x$$

$$\sin\left(\frac{1}{2} \ln x\right) \cdot x - \frac{1}{2} \int \cancel{x} \cos\left(\frac{1}{2} \ln x\right) \frac{1}{\cancel{x}} dx$$

$$\int \cos\left(\frac{1}{2} \ln x\right) dx$$

$$u = \cos\left(\frac{1}{2} \ln x\right) \quad dv = dx$$

$$du = -\sin\left(\frac{1}{2} \ln x\right) \cdot \frac{1}{2} \cdot \frac{1}{x} dx \quad \leftarrow \begin{matrix} \nearrow \\ \nwarrow \end{matrix} \quad v = x$$

$$\cos\left(\frac{1}{2} \ln x\right) \cdot x + \frac{1}{2} \int \cancel{x} \sin\left(\frac{1}{2} \ln x\right) \frac{1}{\cancel{x}} dx$$

$$I = \sin\left(\frac{1}{2} \ln x\right) x - \frac{1}{2} \left[\cos\left(\frac{1}{2} \ln x\right) x + \frac{1}{2} I \right]$$

$$I = \sin\left(\frac{1}{2} \ln x\right) x - \frac{1}{2} \cos\left(\frac{1}{2} \ln x\right) x - \frac{1}{4} I$$

$$I + \frac{1}{4} I$$

$$I \left[1 + \frac{1}{4} \right]$$

$$I \cdot \frac{5}{4}$$

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$$\Rightarrow I = \frac{4}{5} \left[\sin\left(\frac{1}{2} \ln x\right) x - \frac{1}{2} \cos\left(\frac{1}{2} \ln x\right) x \right] + C$$



تکامل الدوال المجهولة



$$\text{if } f(4) = 5, \quad \int_0^4 f(x) dx = 9$$

$$\text{Find } \int_0^4 x f'(x) dx$$

$$\int_0^4 x f'(x) dx$$

$$u = x$$

$$du = dx$$

$$dv = f'(x) dx$$

$$v = f(x)$$

$$\left[x f(x) \right]_0^4 - \int_0^4 f(x) dx$$

$$4 f(4) - 0 - 9$$

$$4 (5) - 9$$

$$20 - 9 = 11$$

If $f(5) = 8$, $f(0) = 0$, $\int_0^5 \sqrt{1+f(x)} dx = 7$

Find $\int_0^5 \frac{x f'(x)}{\sqrt{1+f(x)}} dx$

$$\int_0^5 \frac{x f'(x)}{\sqrt{1+f(x)}} dx$$

$$u = x$$

$$du = dx$$

$$dv = \frac{f'(x)}{\sqrt{1+f(x)}} dx$$

$$v = \int f'(x) (1+f(x))^{-\frac{1}{2}} dx$$

$$\text{Let } z = 1+f(x)$$

$$dz = f'(x) dx$$

$$\int z^{-\frac{1}{2}} dz$$

$$\frac{z^{\frac{1}{2}}}{\frac{1}{2}}$$

$$2\sqrt{z}$$

$$2\sqrt{1+f(x)}$$

$$\left[2x \sqrt{1+f(x)} \right]_0^5 - 2 \int_0^5 \sqrt{1+f(x)} dx$$

$$2(5)\sqrt{1+f(5)} - 0 - 2(7)$$

$$10\sqrt{1+8} - 14$$

$$10(3) - 14 = 30 - 14 = 16$$

if $f(0) = g(0) = 0$, f'' and g'' are continuous

Show that: (using by parts)

$$\int_0^a f(x) g''(x) dx = f(a)g'(a) - f'(a)g(a) + \int_0^a f''(x) g(x) dx$$

$$\text{L.H.S} = \int_0^a f(x) g''(x) dx$$

$$u = f(x) \quad dv = g''(x) dx$$

$$du = f'(x) dx \quad \leftarrow \quad v = g'(x)$$

$$\left[f(x) g'(x) \right]_0^a - \int_0^a g'(x) f'(x) dx$$

$$f(a)g'(a) - 0$$

$$\boxed{f(a)g'(a)}$$

$$u = f'(x)$$

$$dv = g'(x) dx$$

$$du = f''(x) dx \quad \leftarrow \quad v = g(x)$$

$$\left[f'(x) g(x) \right]_0^a - \int_0^a g(x) f''(x) dx$$

$$f'(a)g(a) - 0$$

$$\boxed{f'(a)g(a)}$$

$$\Rightarrow f(a)g'(a) - \left[f'(a)g(a) - \int_0^a g(x) f''(x) dx \right]$$

$$f(a)g'(a) - f'(a)g(a) + \int_0^a g(x) f''(x) dx$$

= R.H.S



Remark



مشتقة التكاملات المحددة

عوض

اشتق

–

عوض

اشتق



Find $f'(x)$ if

$$f(x) = \int_{x^3}^{x^7} t^{10} dt$$

$$f'(x) = (x^7)^{10} \cdot 7x^6 - (x^3)^{10} \cdot 3x^2$$

Find $f'(x)$ if

$$f(x) = \int_{\tan x}^5 (t^2 + 3) dt$$

$$f'(x) = 0 - ((\tan x)^2 + 3) \sec^2 x$$



Trick



$$\text{if } f(x) = \int_5^x \frac{\sin t}{t} dt$$

Find

$$F(x) = \int f(x) dx$$

$$F(x) = \int f(x) dx$$

$$= \int \int_5^x \frac{\sin t}{t} dt dx$$

$$u = \int_5^x \frac{\sin t}{t} dt \quad dv = dx$$

$$du = \left(\frac{\sin x}{x} (1) - 0 \right) dx \quad v = x$$

$$\int_5^x \frac{\sin t}{t} dt \cdot x - \int x \frac{\sin x}{x} dx$$

$$\int_5^x \frac{\sin t}{t} dt \cdot x + \cos x + C$$



عند امكانية تبسيط السؤال قبل حله فانه يتم ذلك

بشرط

(1) وجود دالة ومشتقتها

(2) تحول السؤال بأكمله الى صيغة المتغير الجديد



Text Book Questions



$$\int \frac{\arcsin(\ln x)}{x} dx$$

$$\int \frac{\sin^{-1}(\ln x)}{x} dx$$

$$\text{Let } t = \ln x$$

$$dt = \frac{1}{x} dx$$

$$\int \sin^{-1} t \, dt$$

$$u = \sin^{-1} t \quad dv = dt$$

$$du = \frac{1}{\sqrt{1-t^2}} dt \quad v = t$$

$$\sin^{-1} t \cdot t - \int \frac{t}{\sqrt{1-t^2}} dt$$

$$\sin^{-1} t \cdot t - \int t (1-t^2)^{-\frac{1}{2}} dt$$

$$\text{Let } z = 1-t^2$$

$$dz = -2t dt \rightarrow \frac{1}{2} dz = -t dt$$

$$\sin^{-1} t \cdot t - (-\frac{1}{2}) \int z^{-\frac{1}{2}} dz$$

$$\frac{1}{2} \frac{z^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$\sin^{-1} t \cdot t + \sqrt{1-t^2} + C$$

$$\sin^{-1}(\ln x) \ln x + \sqrt{1-(\ln x)^2} + C$$

$$\int \frac{\sin^{-1}(e^x)}{e^{-x}} dx$$

الزاوية ليست
x'

Let $t = e^x$
 $dt = e^x dx$

$$\int e^x \sin^{-1}(e^x) dx$$

$$\int \sin^{-1} t dt$$

$u = \sin^{-1} t$ $dv = dt$

$du = \frac{1}{\sqrt{1-t^2}} dt$ $v = t$

$$\sin^{-1} t \cdot t - \int \frac{t}{\sqrt{1-t^2}} dt$$

$$\int t (1-t^2)^{-\frac{1}{2}} dt$$

Let $z = 1-t^2$
 $dz = -2t dt$
 $-\frac{1}{2} dz = t dt$

$$-\frac{1}{2} \int z^{-\frac{1}{2}} dz$$

$$-\frac{1}{2} \cdot \frac{z^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$- (1-t^2)^{\frac{1}{2}} + C$$

$$\Rightarrow \sin^{-1} t \cdot t + (1-t^2)^{\frac{1}{2}} + C$$

$$= \sin^{-1} e^x \cdot e^x + (1-(e^x)^2)^{\frac{1}{2}} + C$$

$$\int x^5 e^{-x^3} dx$$

Let $t = x^3$

$$dt = 3x^2 dx \rightarrow \frac{1}{3} dt = x^2 dx$$

$$\int \underline{x^2} x^3 e^{-x^3} \underline{dx}$$

$$\frac{1}{3} \int t e^{-t} dt$$

$$u = t \quad dv = e^{-t} dt$$

$$du = dt \quad v = \frac{e^{-t}}{-1}$$

$$-t e^{-t} + \int e^{-t} dt$$

$$-t e^{-t} + \frac{e^{-t}}{-1} + C$$

$$\Rightarrow \frac{1}{2} [-t e^{-t} - e^{-t}] + C$$

$$\frac{1}{2} [-x^3 e^{-x^3} - e^{-x^3}] + C$$

$$\int \frac{e^{-2x^{-2}}}{x^5} dx$$

Let $t = x^{-2}$

$$dt = -2x^{-3} dx \longrightarrow -\frac{1}{2} dt = x^{-3} dx$$

$$\int x^{-5} e^{-2x^{-2}} dx$$

$$\int \underline{x^{-3}} x^{-2} e^{-2x^{-2}} \underline{dx}$$

$$-\frac{1}{2} \int t e^{-2t} dt$$

$$u = t$$

$$du = dt$$

$$dv = e^{-2t} dt$$

$$v = \underline{\underline{\frac{e^{-2t}}{-2}}}$$

$$= \frac{1}{2} t e^{-2t} + \frac{1}{2} \int e^{-2t} dt$$

$$= \frac{1}{2} t e^{-2t} + \frac{1}{2} \underline{\underline{\frac{e^{-2t}}{-2}}} + C$$

$$\Rightarrow -\frac{1}{2} \left[-\frac{1}{2} t e^{-2t} - \frac{1}{4} e^{-2t} \right] + C$$

$$= \frac{1}{2} \left[\frac{1}{2} x^{-2} e^{-2x^{-2}} - \frac{1}{4} e^{-2x^{-2}} \right] + C$$

$$\int e^{2x} \ln(1 + e^x) dx$$

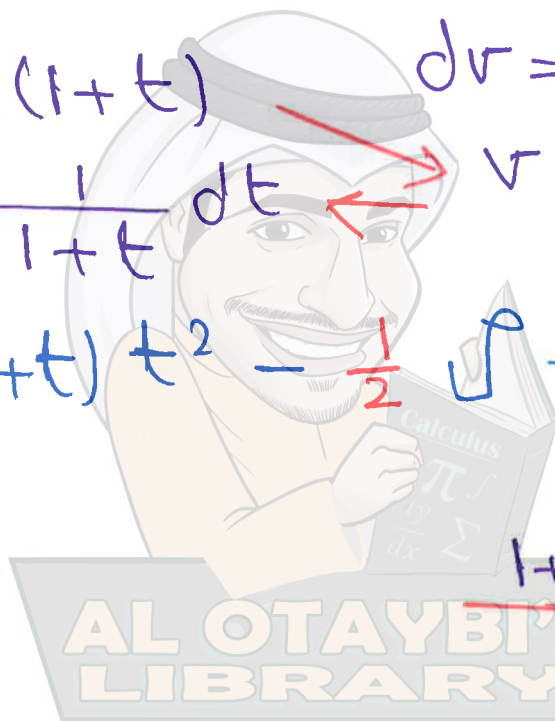
Let $t = e^x$
 $dt = e^x dx$

$$\int \underline{e^x} \underline{e^x} \ln(1 + e^x) \underline{dx}$$

$$\int t \ln(1+t) dt$$

$u = \ln(1+t) \quad dv = t dt$
 $du = \frac{1}{1+t} dt \quad v = \frac{t^2}{2}$

$$\frac{1}{2} \ln(1+t) t^2 - \frac{1}{2} \int \frac{t^2}{1+t} dt$$



$$\begin{array}{r} t-1 \\ 1+t \overline{) t^2} \\ \underline{-(t^2 + t)} \\ -t \end{array}$$

$$\begin{array}{r} -t \\ \underline{+(t + 1)} \\ 1 \end{array}$$



$$\frac{1}{2} \ln(1+t) t^2 - \frac{1}{2} \left(\frac{t^2}{2} - t + \ln|1+t| \right) + C$$

$$\frac{1}{2} \ln(1+e^x) (e^x)^2 - \frac{1}{2} \left(\frac{(e^x)^2}{2} - e^x + \ln|1+e^x| \right) + C$$

$$\int \left[t-1 + \frac{1}{1+t} \right] dt$$

$$\frac{t^2}{2} - t + \ln|1+t| + C$$



Text Book Question



$$\int_0^{\frac{\pi}{3}} \sin x \ln(\cos x) dx$$

Let $t = \cos x$

$$dt = -\sin x dx \rightarrow -dt = \sin x dx$$

$$-\int \ln t dt$$

$$u = \ln t \quad dv = dt$$

$$du = \frac{1}{t} dt \quad v = t$$

$$- \left[\ln t \cdot t - \int t \cdot \frac{1}{t} dt \right]$$

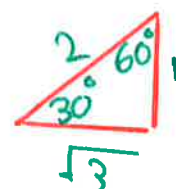
$$- \left[\ln t \cdot t - t \right]$$

$$- \left[\ln \cos x \cdot \cos x - \cos x \right]_0^{\frac{\pi}{3}}$$

$$- \left[\left(\ln \frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \right) - \left(\ln(1) - 1 \right) \right]$$

$$- \left[\frac{1}{2} \ln \frac{1}{2} - \frac{1}{2} + 1 \right]$$

$$- \left[\frac{1}{2} \ln \frac{1}{2} + \frac{1}{2} \right]$$





Text Book Question



$$\int_0^{\pi} e^{\cos t} \sin 2t \, dt$$

Let $t = \cos t$

$$dt = -\sin t \, dt$$

$$-dt = \sin t \, dt$$

$$\int e^{\cos t} 2 \sin t \cos t \, dt$$

$$2 \int e^{\cos t} \boxed{\sin t} \boxed{\cos t} \, dt$$

$$- 2 \int e^t t \, dt$$

$$u = t \quad \begin{matrix} \swarrow \\ du = dt \end{matrix} \quad \begin{matrix} \searrow \\ dv = e^t \, dt \\ v = e^t \end{matrix}$$

$$-2 \left[t e^t - \int e^t \, dt \right]$$

$$-2 \left[t e^t - e^t \right]$$

$$-2 \left[\cos t e^{\cos t} - e^{\cos t} \right]_0^{\pi}$$

$$-2 \left[(-1)e^{-1} - e^{-1} - (1)e^1 - e^1 \right]$$

$$-2 \left[-2e^{-1} \right] = 4e^{-1}$$

$$\int \tan^{-1}(\sinh x) \operatorname{sech}^2 x \, dx$$

الزاوية ليست حدودية من الدرجة الأولى ← راح أفكر بالتبسيط

Let $t = \sinh x$
 $dt = \cosh x \, dx$



المشتقة غير موجودة



فشل التبسيط

$$u = \tan^{-1}(\sinh x)$$

$$dv = \operatorname{sech}^2 x \, dx$$

$$du = \frac{\cosh x \, dx}{(\sinh x)^2 + 1}$$

$$= \frac{\cosh x \, dx}{\cosh^2 x}$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\cosh^2 x = \sinh^2 x + 1$$

$$= \frac{1}{\cosh x} \, dx$$

$$= \operatorname{sech} x \, dx$$

$$\tan^{-1}(\sinh x) \tanh x - \int \tanh x \operatorname{sech} x \, dx$$

$$\tan^{-1}(\sinh x) \tanh x + \operatorname{sech} x + C$$

$$\int \csc x \cot x \ln(\sin x) dx$$

الزاوية ليست حدودية من الدرجة الأولى ← راح أفكر بالتبسيط

Let $t = \sin x$
 $dt = \cos x dx$



المشتقة غير موجودة



فشل التبسيط

$u = \ln(\sin x)$ $dv = \csc x \cot x dx$
 $du = \frac{\cos x}{\sin x} dx$
 $= \cot x dx$ $v = -\csc x$



المشتقة مخشوشة



انظر الحل الثاني

— $\ln(\sin x) \csc x + \int \csc x \cot x dx$
 — $\ln(\sin x) \csc x - \csc x + C$

$$\int \csc x \cot x \ln(\sin x) dx$$

الزاوية ليست حدودية من الدرجة الأولى ← راح أفكر بالتبسيط

Let $t = \sin x$

$dt = \cos x dx$



لو انتبهتي لوجود المشتقة



$$\int \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} \cdot \ln(\sin x) dx$$

$$\int \frac{1}{t} \cdot \frac{1}{t} \ln t dt$$

$$\int t^{-2} \ln t dt$$

$u = \ln t$ $dv = t^{-2} dt$

$du = \frac{1}{t} dt$ $v = \frac{t^{-1}}{-1}$

$$- \ln t \cdot t^{-1} + \int t^{-1} \frac{1}{t} dt$$

$$- \ln t \cdot t^{-1} + \int t^{-2} dt$$

$$- \ln t \cdot t^{-1} + \frac{t^{-1}}{-1} + C$$

$$- \ln t \cdot \frac{1}{t} - \frac{1}{t} + C$$

$$- \ln \sin x \cdot \frac{1}{\sin x} - \frac{1}{\sin x} + C$$

$$\int e^{\sqrt{x}} dx$$



الأس ليس حدودية من الدرجة الأولى

let $t = \sqrt{x}$

$$dt = \frac{1}{2\sqrt{x}} dx$$

$$2dt = \frac{1}{\sqrt{x}} dx$$

$$\int \frac{\sqrt{x}}{\sqrt{x}} e^{\sqrt{x}} dx$$

$$2 \int t e^t dt$$

$$u = t \quad dv = e^t dt$$

$$du = dt \quad v = e^t$$

$$t e^t - \int e^t dt$$

$$t e^t - e^t + C$$

$$\Rightarrow 2 [t e^t - e^t] + C$$

$$2 [\sqrt{x} e^{\sqrt{x}} - e^{\sqrt{x}}] + C$$



ما طلعت المشتقة

ولكن لاحظ بأن المشتقة نفس الفرض



أضرب فيها فوق وتحت

$$\int \tan^{-1} \sqrt{x} dx$$



الزاوية ليست حدودية من الدرجة الأولى

$$t = \sqrt{x}$$

$$dt = \frac{1}{2\sqrt{x}} dx$$

$$2dt = \frac{1}{\sqrt{x}} dx$$



ما طلعت المشتقة
ولكن لاحظ بأن المشتقة نفس الفرض



أضرب فيها فوق وتحت

$$\int \frac{\sqrt{x}}{\sqrt{x}} \tan^{-1} \sqrt{x} dx$$

$$2 \int t \tan^{-1} t dt$$

$$u = \tan^{-1} t \quad dv = t dt$$

$$du = \frac{1}{t^2 + 1} dt \quad v = \frac{t^2}{2}$$

$$\frac{1}{2} \tan^{-1} t \cdot t^2 - \frac{1}{2} \int \frac{t^2}{t^2 + 1} dt$$



$$2 \left[\frac{1}{2} \tan^{-1} t \cdot t^2 - \frac{1}{2} (t - \tan^{-1} t) \right] + C$$

$$2 \left[\frac{1}{2} \tan^{-1} \sqrt{x} (\sqrt{x})^2 - \frac{1}{2} (\sqrt{x} - \tan^{-1} \sqrt{x}) \right] + C$$

$$\frac{1+t^2}{t^2+1} \cdot \frac{t^2}{t^2+1}$$

$$\int \left[1 + \frac{-1}{t^2+1} \right] dt$$

$$t - \tan^{-1} t + C$$

* انبته *

هناك 3 مسائل اضافية في by parts
سيتم سرها بمذكرة 7.3



Prove



Let f and g be two differentiable functions.

Fall 2021

Prove that

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

$\therefore f(x)$ and $g(x)$ are differentiable

$$\therefore \frac{d}{dx} [f(x) \cdot g(x)] = f'(x) \cdot g(x) + f(x)g'(x)$$

Integrate both sides

$$f(x) \cdot g(x) = \int [f'(x) \cdot g(x) + f(x) \cdot g'(x)] dx$$

$$f(x) \cdot g(x) = \int f'(x) \cdot g(x) dx + \int f(x)g'(x) dx$$



$$\therefore \int f(x)g'(x) dx = f(x) \cdot g(x) - \int f'(x)g(x) dx$$



Quiz #2



$$\int \left(\frac{1}{2}x + e^{\frac{1}{2}x} + \left(\frac{1}{2}\right)^x \right) dx$$

$$= \frac{1}{2} \frac{x^2}{2} + \frac{e^{\frac{1}{2}x}}{\frac{1}{2}} + \frac{\left(\frac{1}{2}\right)^x}{\ln \frac{1}{2}} + C$$

$$\int \frac{2}{\sqrt{9-x^2}} dx$$

$$2 \sin^{-1}\left(-\frac{x}{3}\right) + C$$

$$\int \frac{2x}{\sqrt{9-x^2}} dx$$

$$\int 2x(9-x^2)^{-\frac{1}{2}} dx$$

$$\begin{aligned} \text{Let } u &= 9-x^2 \\ du &= -2x dx \\ -du &= 2x dx \end{aligned}$$

$$\begin{aligned} &= -\int u^{-\frac{1}{2}} du \\ &= \frac{-u^{\frac{1}{2}}}{\frac{1}{2}} + C \\ &= -2(9-x^2)^{\frac{1}{2}} + C \end{aligned}$$

$$\int \frac{3^{\ln x}}{x} dx$$

$$\begin{aligned} \text{Let } u &= \ln x \\ du &= \frac{1}{x} dx \end{aligned}$$

$$\begin{aligned} &\int 3^u du \\ &= \frac{3^u}{\ln 3} + C = \frac{3^{\ln x}}{\ln 3} + C \end{aligned}$$

$$\int x^2 \cot(x^3 + 2) dx$$

$$\begin{aligned} \text{Let } u &= x^3 \\ du &= 3x^2 dx \longrightarrow \frac{1}{3} du = x^2 dx \end{aligned}$$

$$\begin{aligned} &\frac{1}{3} \int \cot(u+2) du \\ &\frac{1}{3} \ln |\sin(u+2)| + C \\ &\frac{1}{3} \ln |\sin(x^3+2)| + C \end{aligned}$$

$$\int \left(\frac{\sinh x}{\cosh x + 3} - \frac{2\sinh x}{\cosh x + 3} + \frac{\sinh x}{2\cosh x + 3} \right) dx$$

$$\int \left(\frac{\sinh x}{\cosh x + 3} - \frac{2\sinh x}{\cosh x + 3} + \frac{1}{2} \frac{2\sinh x}{2\cosh x + 3} \right) dx$$

$$= \ln |\cosh x + 3| - 2 \ln |\cosh x + 3| + \frac{1}{2} \ln |2\cosh x + 3| + C$$

$$\int \frac{\sinh x}{\cosh^2 x + 3} dx$$

$$\int \frac{\sinh x}{(\cosh x)^2 + 3} dx$$

$$\begin{aligned} u &= \cosh x \\ du &= \sinh x dx \end{aligned}$$

$$\int \frac{1}{u^2 + 3} du$$

$$= \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{u}{\sqrt{3}}\right) + C$$

$$= \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{\cosh x}{\sqrt{3}}\right) + C$$